ETH zürich



Introduction to Scientific Computation
Lecture 4
Fall 2026

Eigenvalues, PageRank, ODEs

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Eigenvalues and PageRank

See `lecture_code.ipynb`

Ordinary Differential Equations (ODEs)

An equation is called *differential* if it includes derivatives of an unknown function.

A differential equation whose all derivatives are with respect to a single argument is called *ordinary*. The order of the ODE is the highest derivative involved, e.g. a first-order ODE can be:

$$\frac{dy}{dx} = f(x, y)\,.$$

Solving an ODE is sometimes called *integration*, and numerically also *simulation*.

Higher-Order ODEs

A higher-order ordinary differential equation can be reduced to a system of first-order ones by introducing explicit variables for the intermediate derivatives:

$$y_0 = y;\ y_1 = \frac{dy_0}{dx} = f(x, y_0);\ y_2 = \frac{dy_1}{dx} = \frac{d^2y}{dx};\ \ldots;\ y_n = \frac{dy_n}{dx} = \frac{d^ny}{dx^n}$$

Solving ODEs Numerically

Numerical solutions are sought for an Initial value problem:

$$\frac{dy}{dt} = f(t, y),\ y(t_0) = y_0$$

over discrete grid points

$$t_{n+1} = t_n + h_n\,.$$

Solving ODEs Numerically: Euler's Methods

Explicit Euler (Forward):

$$y_{n+1} = y_n + hf(t_n, y_n)\,.$$

It is a computationally cheap with *O(h)* accuracy but very limited stability domain.

Implicit Euler (Backward):

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1})\,.$$

An unconditionally stable method with *O(h)* accuracy but requires solving a nonlinear system at each step.

Solving ODEs Numerically: Runge-Kutta

Classical RK4:

$$y_{n+1} = y_n + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right),$$
$$t_{n+1} = t_n + h$$

with

$$k_1 = f(t_n, y_n),$$
$$k_2 = f\left(t_n + \frac{h}{2}, y_n + k_1\frac{h}{2}\right),$$
$$k_3 = f\left(t_n + \frac{h}{2}, y_n + k_2\frac{h}{2}\right),$$
$$k_4 = f\left(t_n + h, y_n + hk_3\right).$$

It achieves $O(h^4)$ accuracy but still suffers from bounded stability.

It is still an explicit method unsuitable for solving so-called stiff equations [1]. It can be made implicit at a cost of requiring to solve systems of nonlinear equations at every time step.

[1] https://en.wikipedia.org/wiki/Stiff_equation

Solving ODEs Numerically: Some Other Methods

Linear Multistep Methods (LMMs) utilize historical data to avoid multiple intermediate function evaluations. Require a bootstrap of the history using explicit methods.

Symplectic integrators preserve physical invariants (e.g. total system energy) that are otherwise drifting in standard integrators.